Contains Duplicate I

$-10^9 \le arr_i \le 10^9$

Contains Duplicate I

Given a array of **N** integers values, design an algorithm to **check** if there **exists** a **duplicate** value in the given array or not.

Example

0	1	2	3
1	2	3	1

true

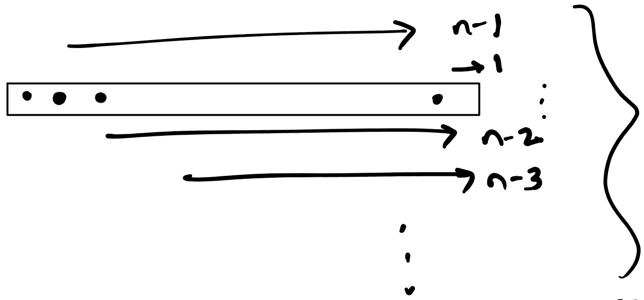
0	1	2	3
1	2	3	4

false

1 → 1
2 → 1
3 → 1
4 → 1

1.

1 → ~~1~~ 2
2 → 1
3 → 1



$O(n^2)$

2. sort + scan

$n\log n + \cancel{n}$ → $O(n\log n)$

3. use freqmap

map< $\underbrace{\text{int}}_{\text{key}}$, $\underbrace{\text{int}}_{\text{value}}$ >
or
unordered_map

key ↑ elements of array

value ↑ freq. of element

Contains Duplicate II

Given a array of **N** integers values and an integer **K**, design an algorithm to **check** if there **exists two distinct** indices **i** and **j** such that values at the **i^{th}** and **j^{th}** indices of the given array are equal and **abs(i - j) <= K**.

Example

K = 3

0	1	2	3
1	2	3	1

K = 2

0	1	2	3	4	5
1	2	3	1	2	3

Longest Palindrome

Given a string which consists of lowercase or uppercase letters, design an algorithm to find the **length** of the **longest palindrome** that can be build with those letters.

Example

Input : "abccccdd"
Output : 7

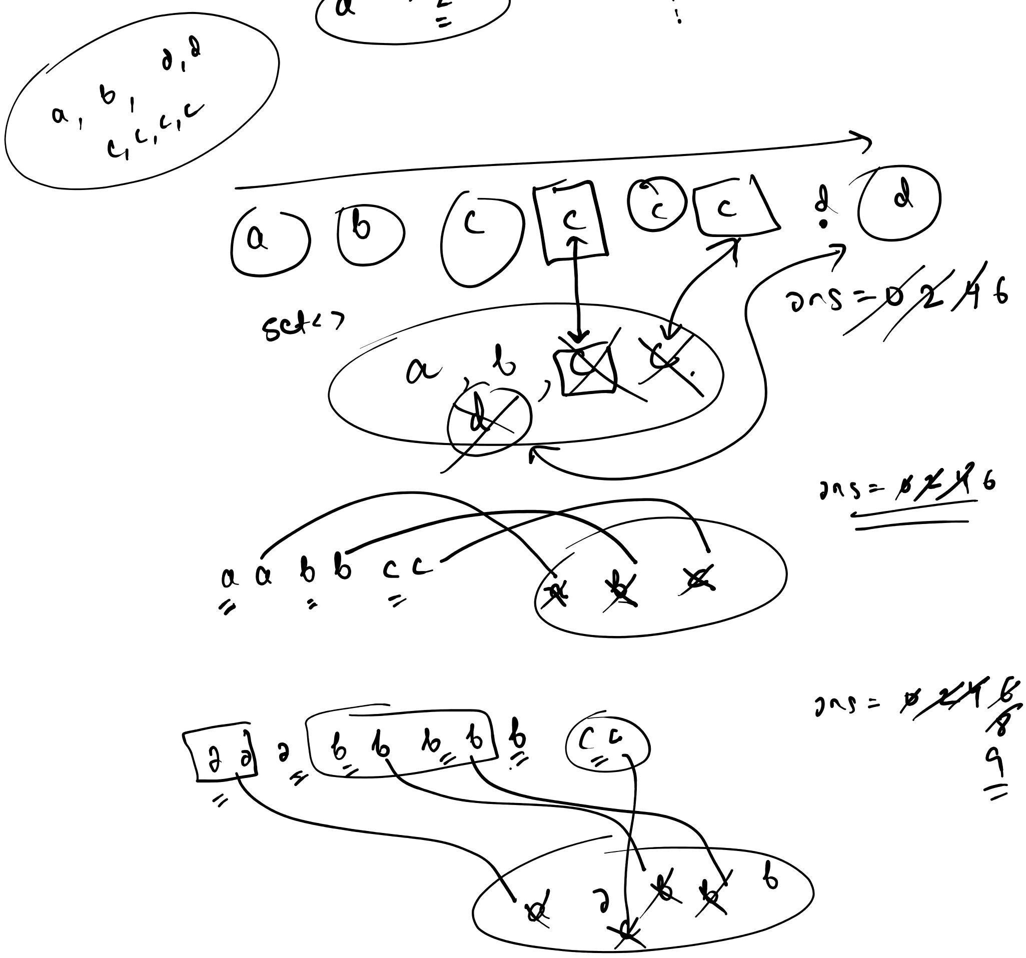
a → 1
b → 1
c → 4
d → 2

d c c b c c d

d c c a c c d

⇒ a → ~~5~~ 1
b → 6
c → 4
⇒ d → ~~7~~ 1

$\underbrace{6bs + 4cs + 6ds + 4as}_{20}$



Longest Consecutive Sequence

cnt = ~~0 1 2 3 4~~ ~~0 1 2 3~~ ... maxSofar = ~~0~~ 4

Given an unsorted array of integers **nums**, design an algorithm to find the length of the **longest** sequence of **consecutive** elements.

Example

0	1	2	3	4	5	6	7	8
100	4	200	1	3	2	101	201	102

Output : 4

0	1	2	3	4	5	6	7	8	9
0	3	7	2	5	8	4	6	0	1

Output : 9

$\underbrace{1\ 2\ 3\ 4}_{4}$ $\underbrace{100\ 101\ 102}_{3}$ $\underbrace{200\ 201}_{2}$

max = 4

time :
nlogn (sorting)
+
n

∴ O(nlogn)

1
100
200

4 1 3 7 2 6 10 9 8 12 13 11

$\boxed{4}$ → ~~true~~ false
$\boxed{1}$ → true
$\boxed{3}$ → ~~true~~ false
7 → ~~true~~ false
$\boxed{2}$ → false

$\boxed{6 \to \text{true}}$
→ 10 → ~~true~~ false
→ 9 → ~~true~~ false
→ 8 → false
→ 12 → ~~true~~ false
→ 13 → false
→ 11 → false

1 2 3 4

6 7 8 9 10 11 12 13

Key = 1

cnt = ~~0 1 2 3 4~~ 0

maxsofar = ~~0~~ ~~4~~ 8

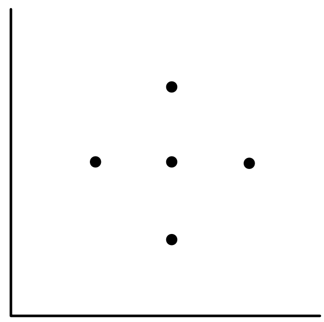
Key = 6
cnt = ~~0 1 2~~
~~3 4~~
~~5~~ 6
~~7~~ 8

Count Triangles

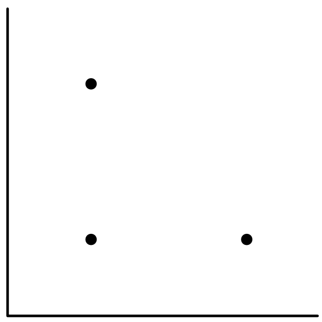
Given **N** cartesian points in a 2D plane, **count** the number of **right-angled** **triangles** that can be formed whose base or perpendicular is parallel to the x-axis or the y-axis.

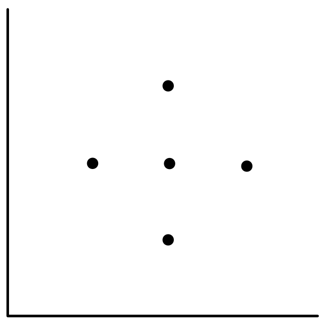
Example :

Input : [(1, 2), (2, 1), (2, 2), (2, 3), (3, 2)]



Output : 4





Input : [(1, 2), (2, 1), (2, 2), (2, 3), (3, 2)]

x-coordinate	frequency

xFreqMap

y-coordinate	frequency

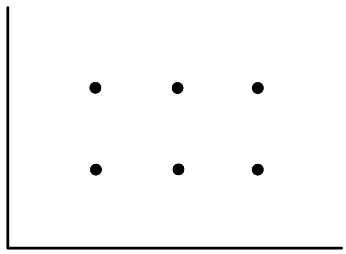
yFreqMap

Count Rectangles

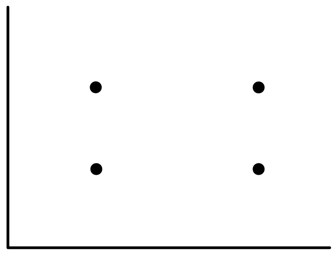
Given **N** cartesian points in a 2D plane, design an algorithm to **count** the number of **axis-parallel rectangles** that can be formed from the given set of points.

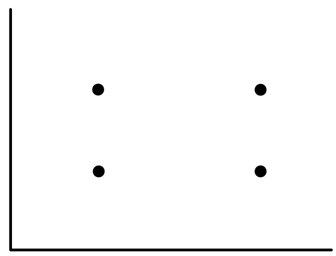
Example :

Input : [(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)]



Output : 3





Find the Duplicate Number I

Given a array of integers containing **N+1** integers where each integer is in the range **[1, N]** **inclusive**. Design an algorithm to find the **duplicate** number given ~~that~~ there is only **one repeated number.**

time: $O(n)$
space: $O(1)$
requirement

Example

0	1	2	3	4
1	3	4	2	2

$n+1 = 5$
$\therefore n = 4$
$x \in [1, 4]$

0	1	2	3	4
3	1	3	4	2

$n+1 = 5$
$\therefore n = 4$
$x \in [1, 4]$

2	2	2	2	2

$n+1 = 5$
$\therefore n = 4$
$x \in [1, 4]$

✓ size = $n+1$ ⇒ idx $\in [0, n]$

✓ elements $\in [1, n]$ ⇒ zero is not present

0	1	2	3	4
1	3	4	2	2

1 2 3 4

0	1	2	3	4
3	1	3	4	2

1 3

0	1	2	3	4
~~1~~	~~3~~	~~4~~	~~2~~	~~2~~

~~3~~ 1 2 3 4
~~2~~
~~4~~
$\boxed{2}$

[HW] Find the Duplicate Numbers II

Find the Duplicate Number II

Given a array of integers containing **N** integers where each integer is in the range **[0, N-1]** **inclusive**. Find all the **duplicate** numbers present in the given array in constant space.

Example

0	1	2	3	4
1	0	1	0	2

0	1	2	3	4
1	2	2	2	3

Homework

https://cses.fi/problemset/task/1661
[R] https://leetcode.com/problems/find-duplicate-subtrees/
[R] https://leetcode.com/problems/vertical-order-traversal-of-a-binary-tree/